

Project Title: Terraform-based Infrastructure Deployment on AWS

Abstract

Problem Statement

Traditional IT systems require manual monitoring, which can cause downtime and poor performance. An automated solution is needed for continuous availability and efficient resource management.

Technology Used

This project uses **Amazon Web Services (AWS)** services such as **Amazon EC2** for hosting applications, **Elastic Load Balancing (ELB)** for traffic distribution, **Auto Scaling** for automatic resource management, and **Amazon CloudWatch** for monitoring system health and performance.

Purpose of the Project

- To build a self-healing cloud infrastructure using AWS.
- To automatically monitor system health, replace failed EC2 instances, and scale resources based on traffic demand.

Expected Outcome

- Provides a highly available and scalable cloud infrastructure.
- Automatically recovers from failures and scales resources.
- Improves performance, reduces downtime, and optimizes cloud costs.

Introduction

1. Cloud computing provides scalable and reliable infrastructure for hosting modern applications.
2. Traditional systems require manual monitoring, which can lead to downtime and reduced performance.
3. This project uses AWS services such as Amazon EC2, Elastic Load Balancing, Auto Scaling, and CloudWatch to automate monitoring and recovery.
4. The system ensures high availability, better performance, and cost efficiency through self-healing and automatic scaling.

Background and Importance of the Project

1. Cloud computing provides reliable and scalable infrastructure for modern applications.
2. Manual monitoring can cause downtime and reduced system performance.

3. This project uses AWS to automate monitoring, recovery, and scaling for better availability and cost efficiency.

Key Points of the Project

- **Amazon EC2** hosts the application.
- **Auto Scaling** adjusts resources automatically.
- **CloudWatch** monitors system health.
- **Self-Healing** replaces failed instances automatically.
- Ensures **high availability** and **cost efficiency**.

Advantages

- Ensures high availability of applications.
- Automatically replaces failed instances.
- Improves system performance during traffic changes.
- Reduces operational costs through auto scaling.

Limitations

- Depends on AWS cloud services.
- Initial setup and configuration can be complex.
- Incorrect scaling settings may increase costs.
- Requires internet connectivity to access cloud resources.

Future Scope

- Add AI-based predictive scaling.
- Improve cloud security.
- Support multi-region deployment.

Conclusion

- The project provides a reliable and scalable cloud infrastructure using AWS.
- It automatically monitors, heals, and scales resources based on demand.
- It improves system availability, performance, and cost efficiency.
- It reduces manual effort and ensures continuous application availability.